PHYS 7501:
Particles and Nuclei I

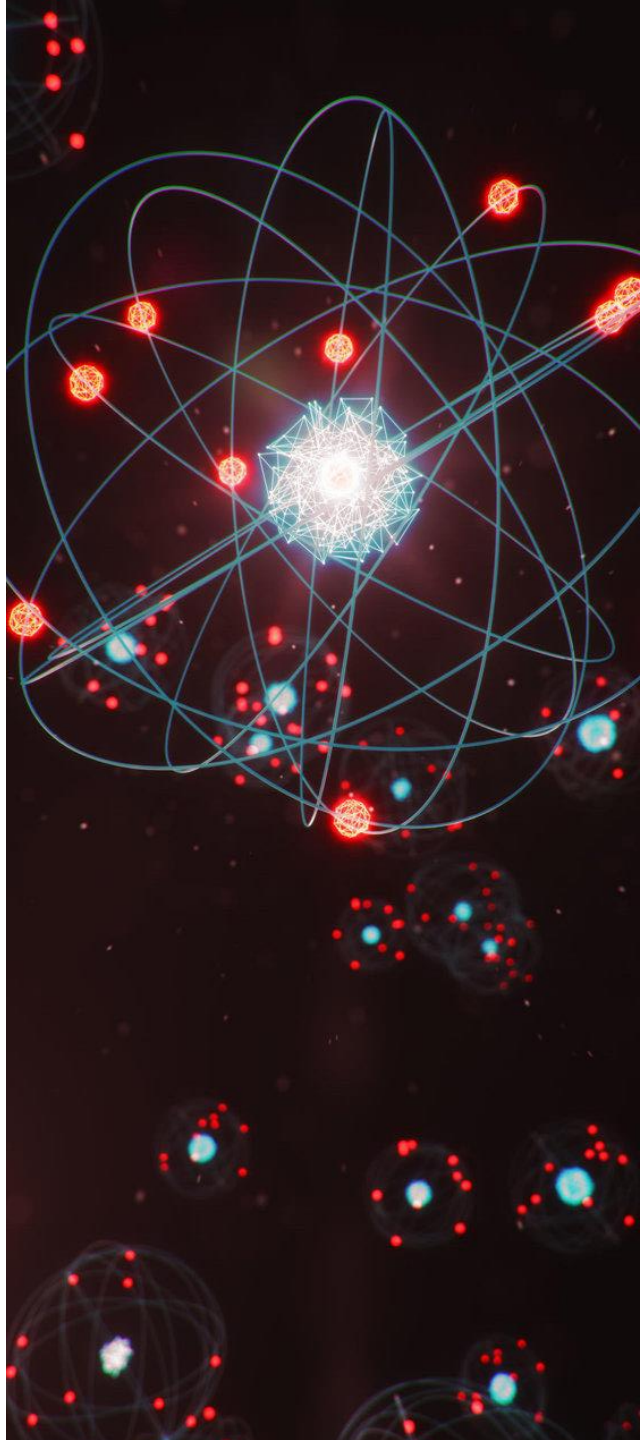
Tues./Thurs. 2 – 3:20PM
Clippinger 105

Dr. Andrea Richard

Edwards Accelerator Lab 204

richarda1@ohio.edu

Available via e-mail & Teams



STATISTICAL REACTIONS

Modern Nuclear Chemistry, Chapter 10

Introductory Nuclear Physics, Krane, Chapter 11

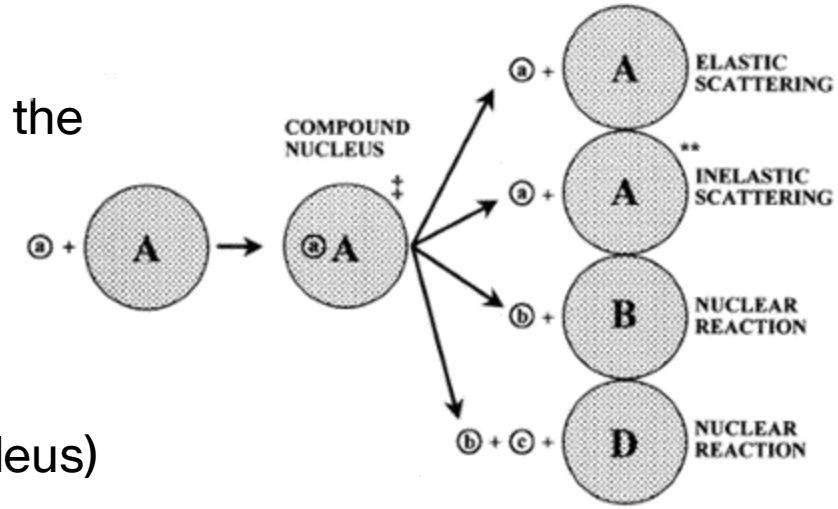
Monday	Tuesday	Wednesday	Thursday	Friday
10/27	10/28	10/29	10/30	10/31
	Statistical Reactions		**Other Reaction Types**	**Topic + Resources due**
11/3	11/4	11/5	11/6	11/7
	Nuclear Astrophysics		**Nuclear Astrophysics**	
11/10	11/11	11/12	11/13	11/14
	No Class		**Nuclear Astrophysics** **HW#4 Due**	

The Semi-Classical View

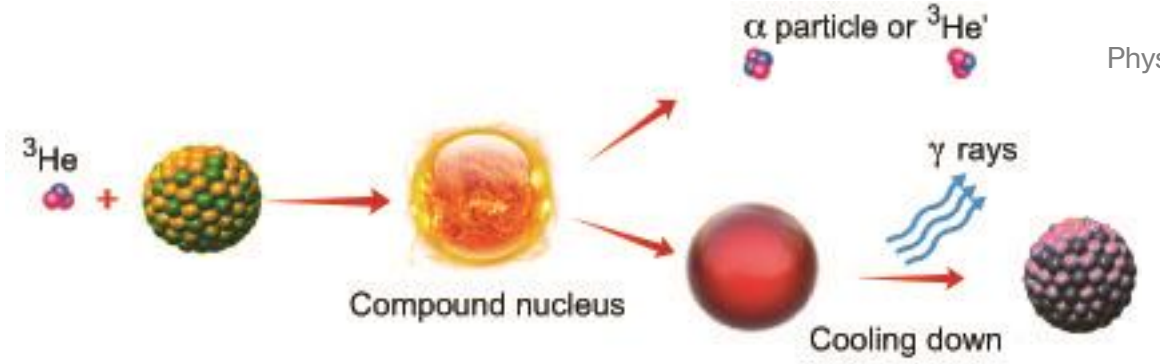
- Consider an incident particle entering a target nucleus with an impact parameter (b) that is small compared to the nuclear radius (i.e. $b \ll R$)
- The particle has a high probability of interacting with one of the nucleons in the target (e.g. via scattering)
- The struck nucleon and incident particle (with less energy) can each make successive collisions with other nucleons, and after several such interactions, the incident energy is shared among many of the nucleons of the combined system of projectile + target
- In other words, the particle will have a high probability of fusing with the target, sharing its energy amongst many nucleons in the nucleus
- The average increase in energy of any single nucleon is not enough to free it from the nucleus, but as many more-or-less random collisions occur, there is a statistical distribution in energies and a small probability for a single nucleon to gain a large enough share of the energy to escape, similar to how molecules evaporate from a hot liquid, often called particle evaporation.
- Such reactions have an intermediate state known as a **compound nucleus**

Compound Nucleus

- Involves the process of forming a relatively long-lived intermediate nucleus and its subsequent decay
- The intermediate state can be written as $\boldsymbol{a + X \rightarrow C* \rightarrow Y + b}$ where $\boldsymbol{C*}$ is the compound nucleus (CN)
- Compound nuclear reactions occur via two steps: formation of the CN and decay of the CN
- What does long-lived mean in this context?
  - Recall our discussion of crossing time (time interval for bombarding particle to travel the diameter of the target nucleus)
  - Here the formation of the CN is $\Delta \boldsymbol{t} \sim \frac{\boldsymbol{R}}{\boldsymbol{c}} \sim \boldsymbol{10^{-21} s}$
  - Then, after another relatively long time period ($\sim 10^{-19} - 10^{-15}$s) and independent of the properties of the reactants, the CN disintegrates



CN Formation and Decay

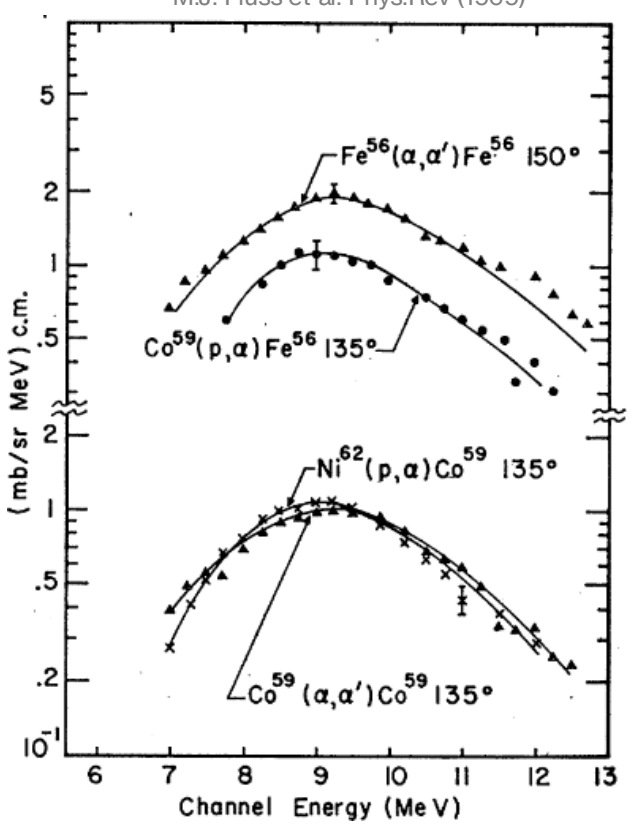


- A given compound nucleus may decay in a variety of different ways, and essential to the compound-nucleus model of nuclear reactions is the assumption that *the relative probability for decay into any specific set of final products is independent of the means of formation of the compound nucleus.* This is the **Bohr Independence Hypothesis** or **the Independence Hypothesis** or **Amnesia Assumption**.
- The decay probability depends only on the total energy given to the system; in effect, the compound nucleus "forgets" the process of formation and decays governed primarily by statistical rules
- The de-excitation of the compound nucleus is akin to evaporation, meaning the ejectile energy distribution should have a Maxwell-Boltzmann character
- The multiple collisions occurring with the nucleus erases any signatures left by the initial reaction, so
  - The ejectiles should be isotropic (in the center of mass frame, since momentum still has to be conserved)
  - The de-excitation characteristics for a given compound nucleus excited state energy (e.g. the ejectiles and their energy distributions) shouldn't depend on how the compound nucleus was created

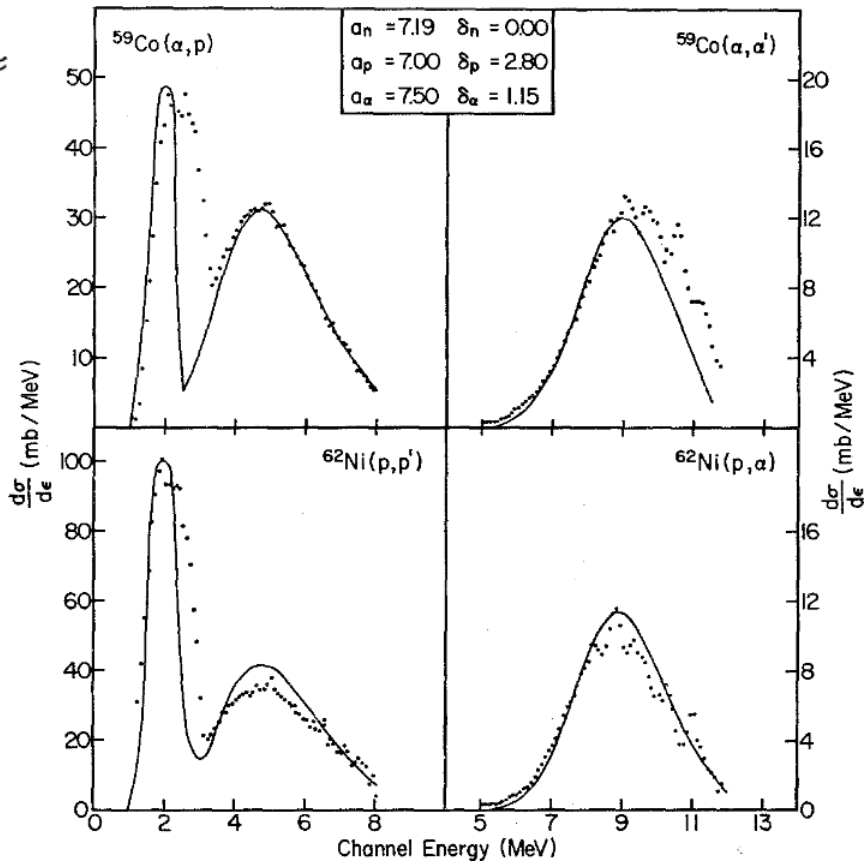
The Independence Hypothesis

- This statistical reaction picture implies that the CN forgets how it was formed, so the decay properties depend only on the CN itself
- This is confirmed by decay spectra for nuclei populated by various channels
- The key implication here is that CN formation and decay probabilities are separated
- One key result which follows is that other probes, e.g. β-decay, can be used to determine key properties needed to understand much harder to measure reactions, e.g. (p,γ)

M.J. Fluss et al. Phys.Rev (1969)

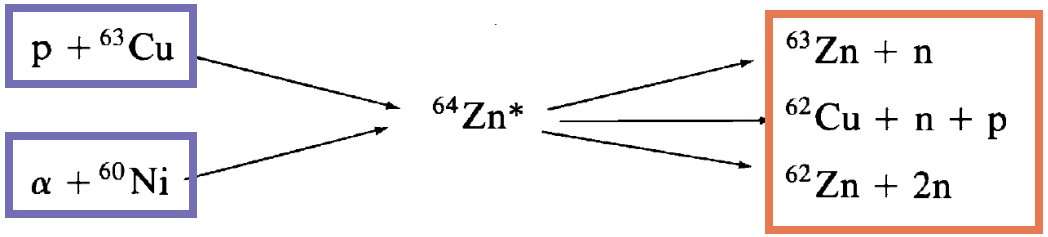


J. Wiley et al., Nuc.Phys.A (1973)

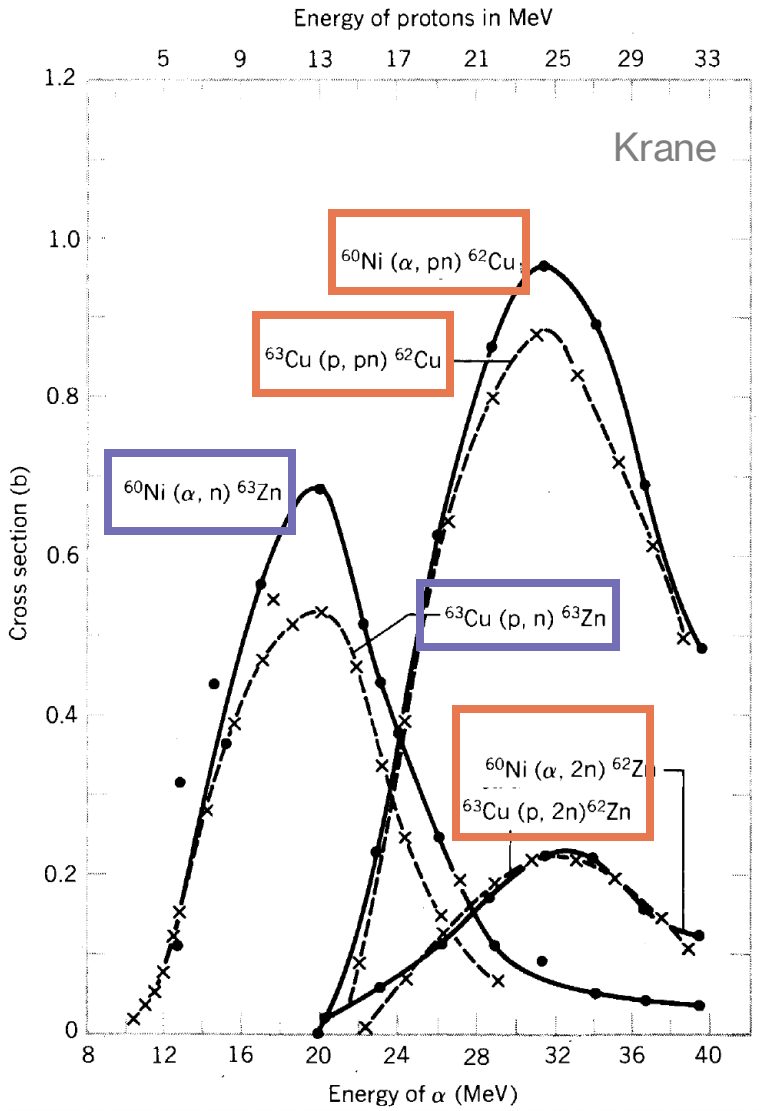


Example of the Independence Hypothesis

- In 1950, S. N. Goshal studied the formation of ^{64}Zn via multiple formation and decay channels, (S. N. Goshal Phys. Rev. **80**, 939 (1950).)



Remarkable agreement between formation and decay channels.



Hauser Feshbach Formalism

- Building on our last lecture about resonances, within this statistical approach (many successive collisions), what we obtain are many resonances that are closely spaced such that the projectile will experience an interaction that's the statistical average of said resonances
- The formalism we will be using is called Hauser-Feshbach Formalism (actually it should be **Wolfenstein-Hauser-Feshbach** (WHF)) but its commonly called HF formalism or HF theory
- Since CN reactions require two steps: formation & decay
- We can write the formation of the CN as $\boldsymbol{\sigma_{CN}} = \left(\frac{\boldsymbol{\lambda}}{\boldsymbol{2\pi}}\right)^{\boldsymbol{2}} \sum_{\ell}(\boldsymbol{2\ell + 1})\boldsymbol{T}_{\ell}$, where $\boldsymbol{T}_{\ell}$ is the transmission coefficient
- The probability of decay of the CN into a given set of products β is:

$$\boldsymbol{P(\beta)} = \left[\frac{\boldsymbol{T_{\ell}^{\beta}(E_{\beta})}}{\sum_{\ell,\boldsymbol{E_i}} \boldsymbol{T_{\ell}^{i}(E_i)}}\right]$$

$\boldsymbol{T_{\ell}^{i}(E_i)}$ is the transmission coefficient for CN decay into products i with some energy $(\boldsymbol{E_i})$

The total cross section then becomes $\sigma_{aX} = \sigma_{CN}\ (aX) \sum P_{CN}(bY)$

Hauser Feshbach Formalism

- Recall from the last lecture that for a single resonance, we have

$$\sigma_{BW,X(a,b)Y}(E) = \pi\left(\frac{\lambda}{\pi}\right)^2 \frac{2J+1}{(2J_a+1)(2J_X+1)} \frac{\Gamma_{aX}(E)\Gamma_{bY}(E)}{(E-E_R)^2 + (\Gamma(E))^2/4}$$

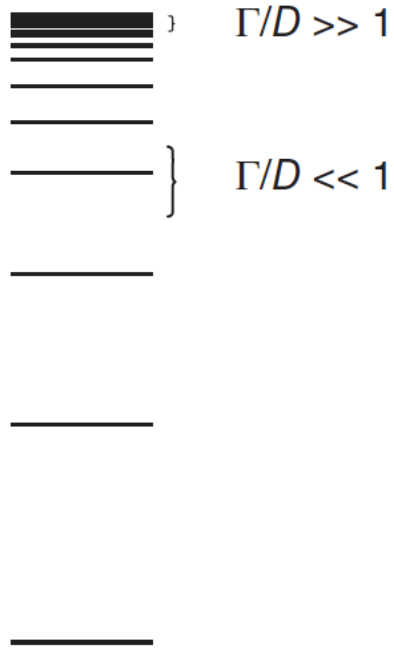
- Averaging over several resonances in an energy range requires integrating over some energy window dE and normalizing by the level-spacing D.

- The average cross section involving compound nuclear resonances for states with spin J is:

$$\left\langle \sigma^J_{X(a,b)Y}(E) \right\rangle = \pi\left(\frac{\lambda}{\pi}\right)^2 \frac{2J+1}{(2J_a+1)(2J_X+1)} \frac{2\pi}{D} \left\langle \frac{\Gamma_{aX}\Gamma_{bY}}{\Gamma} \right\rangle$$

- We would rather be dealing with average widths, rather than averages of their products and ratios, so we take into account their correlations with a width fluctuation factor W_{ab},

$$\left\langle \frac{\Gamma_{aX}\Gamma_{bY}}{\Gamma} \right\rangle = W_{ab} \frac{\langle \Gamma_{aX} \rangle \langle \Gamma_{bY} \rangle}{\langle \Gamma \rangle}$$



HF assumes the level spacing $D \gg \Gamma$

Calculating W_{ab} is actually pretty challenging but is $W_{ab} \approx 1$ when multiple exit channels are open. This is often the case.

Hauser Feshbach Formalism

- The total cross section can then be written

$$\sigma_{X(a,b)Y} = \sum_J \left\langle \sigma^J_{X(a,b)Y}(E) \right\rangle = \pi \left(\frac{\lambda}{\pi}\right)^2 \sum_J \frac{2J+1}{(2J_a+1)(2J_X+1)} \frac{2\pi}{D} W_{ab} \frac{\langle \Gamma_{aX} \rangle \langle \Gamma_{bY} \rangle}{\langle \Gamma \rangle}$$

- This corresponds to the formation and probability of decay of the CN treating it as a sum over all resonances.
- It turns out that we don't (in practice) have all of this information about levels within the CN, so instead we consider the reaction cross section (summing over all exit channels) for the compound nuclear states of spin J as

$$\sigma_{X+a}(E) = \sum_b \sigma_{X(a,b)}(E) = \pi \left(\frac{\lambda}{\pi}\right)^2 \frac{2J+1}{(2J_a+1)(2J_X+1)} \frac{2\pi}{D} \langle \Gamma_{aX} \rangle$$

- If you recall from the Reactions lecture, the reaction cross section for a given orbital angular momentum ℓ is $\sigma_\ell = \pi \left(\frac{\lambda}{\pi}\right)^2 \sum_\ell (2\ell+1) T_\ell$ with T_ℓ being the transmission coefficient for the entrance channel calculated using the optical model potential. $(2\ell+1)$ is for a spinless particle.

Hauser Feshbach Formalism

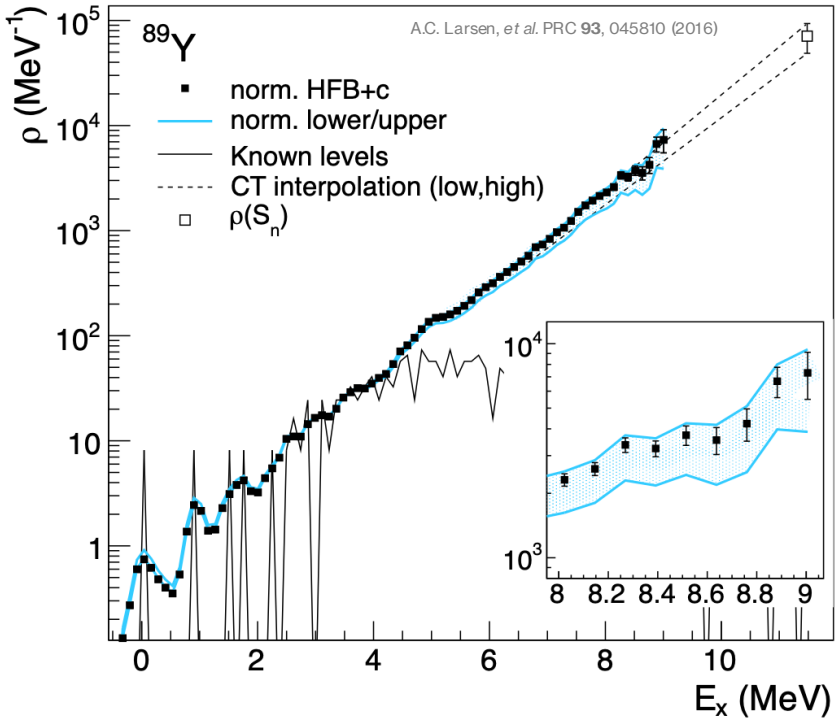
- The transmission coefficient for some reaction channel is related to that channel's average width by $\boldsymbol{T_{chan} = 2\pi \frac{\langle \Gamma_{chan} \rangle}{D}}$
- Since $\boldsymbol{0 \leq T_{chan} \leq 1}$, we can Taylor expand this expression to obtain $\boldsymbol{T_{chan} = 1 - exp\left(-2\pi \frac{\langle \Gamma_{chan} \rangle}{D}\right)}$ so that any choice of $\frac{\langle \Gamma_{chan} \rangle}{D}$ will result in a reasonable value for the transmission coefficient.
- So, swapping in the transmission coefficients yields:

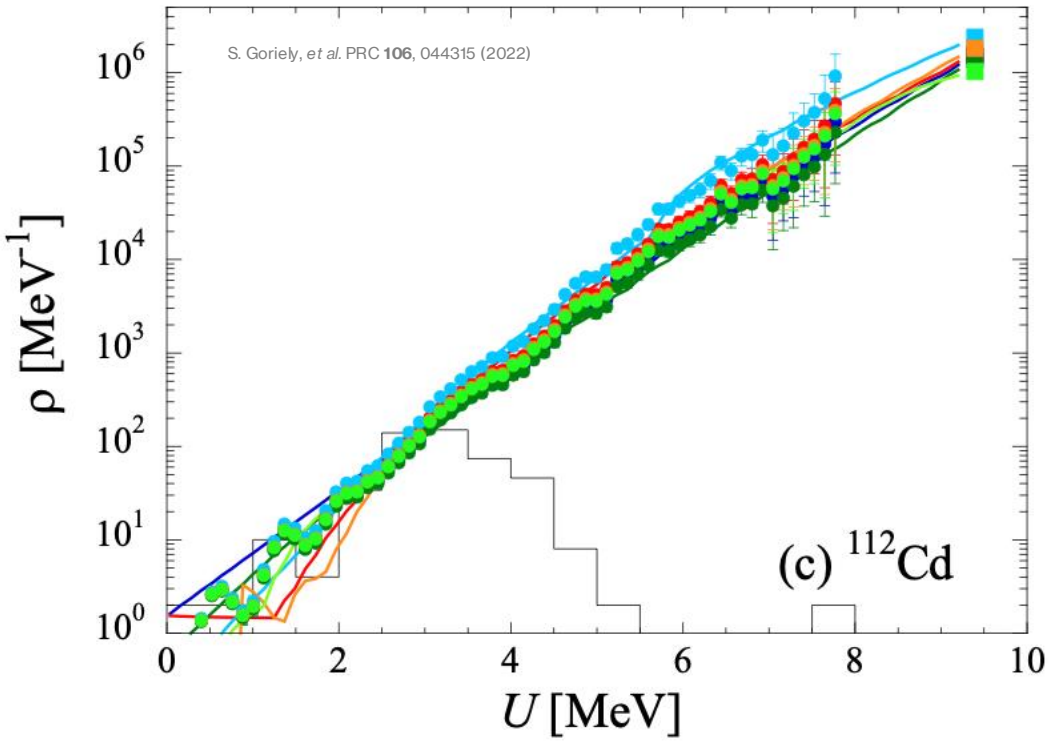
$$\boldsymbol{\sigma_{X(a,b)Y} = \sum_J \left\langle \sigma^J_{X(a,b)Y}(E) \right\rangle = \pi \left(\frac{\lambda}{\pi}\right)^2 \sum_J \frac{2J+1}{(2J_a+1)(2J_X+1)} W_{ab} \frac{T_{aX} T_{aY}}{\sum_{chan} T_{chan}}}$$

- This full sum requires considering angular momentum conservation, parity conservation, and energy conservation (to determine which outgoing channels are possible)
- So, the key ingredients in the HF cross section are
  - The transmission coefficient for the entrance channel — From an optical potential
  - The transmission coefficients for all exit channels — From optical potential
  - The number of available levels (and their energies) for all exit channels — From level density models and spin cut-off parameters

Aside on Level Density

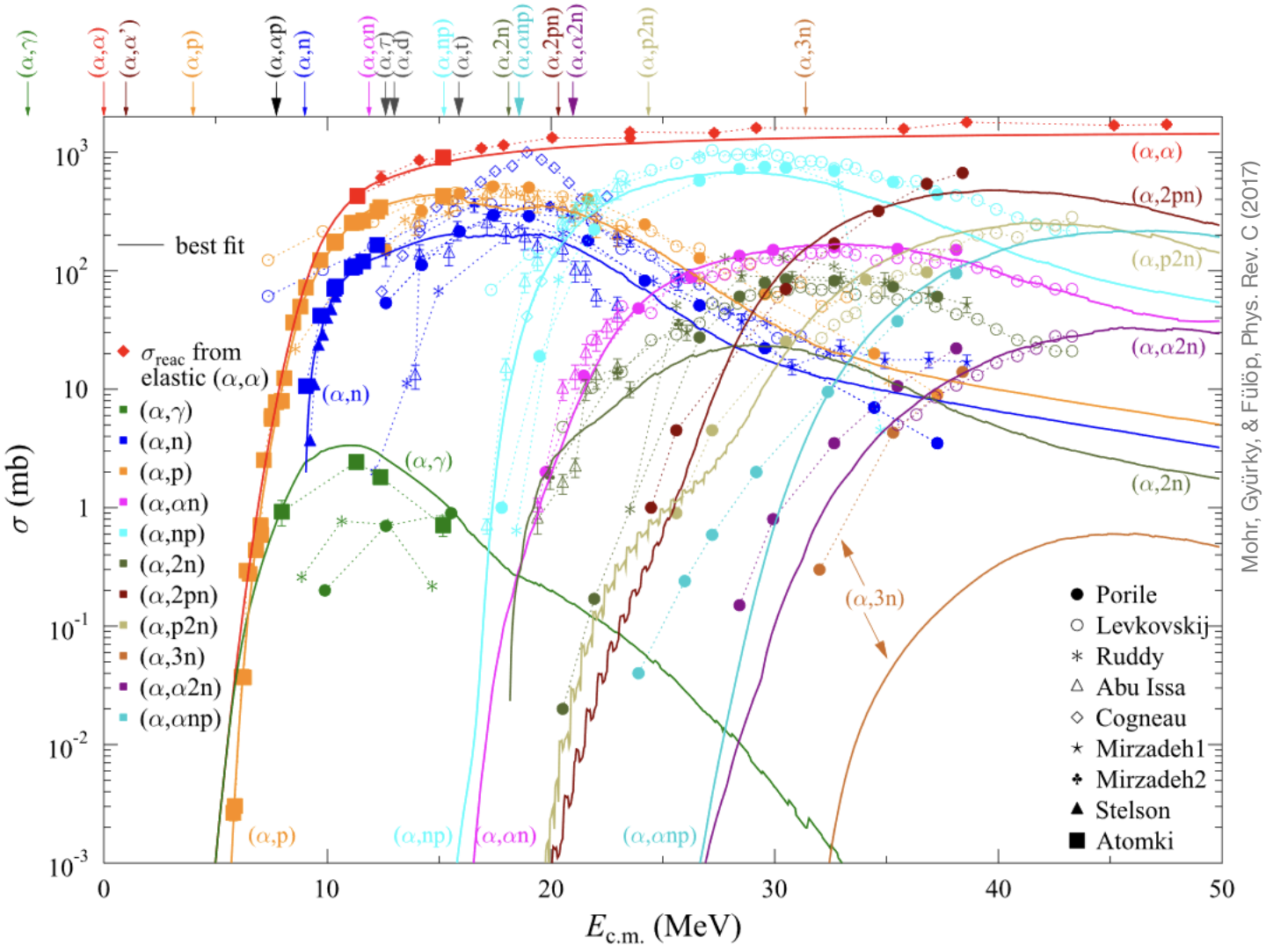
- We know HF formalism relies on knowledge of the available levels (and their energies) for all exit channels – we refer to this as a level density
- A nuclear level density describes the number of available energy levels as a function of excitation energy in the nucleus, which increases rapidly as excitation energy increases.
- Expressed as $\rho(\mathrm{E}*, \mathrm{J})$ for which there are several models (e.g. Fermi Gas Model)





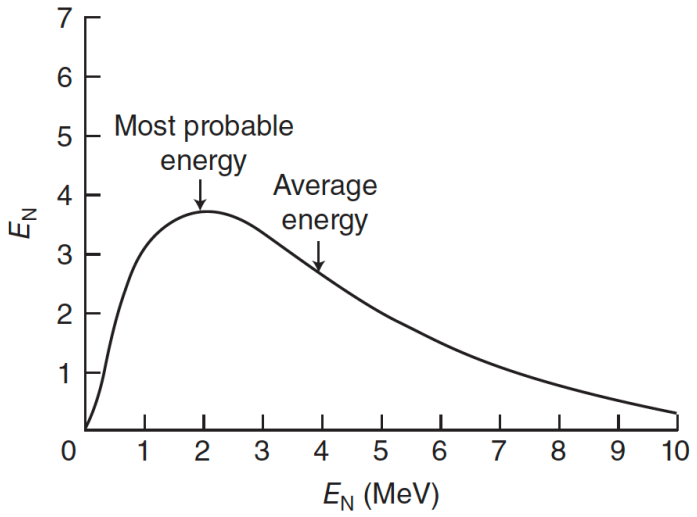
Comparison to Cross Section Measurements

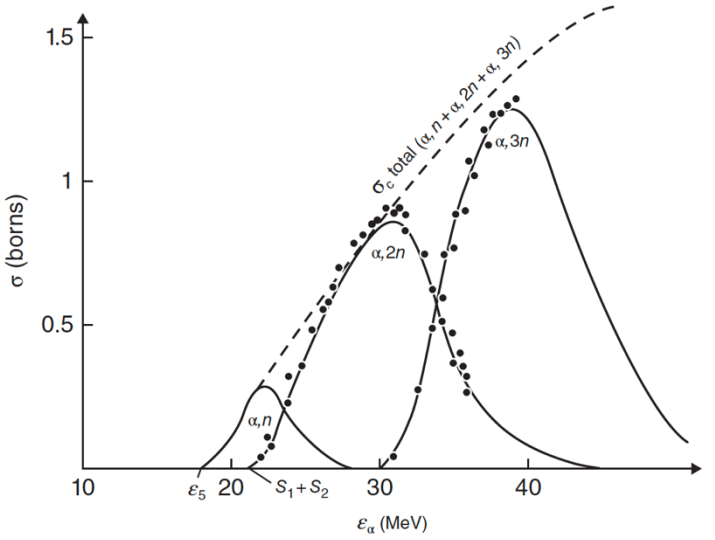
- HF calculations do a great job of reproducing data, where applicable
- Given reasonable input choices, HF predictions can vary by a factor of a few, but for radioactive nuclei where no data is available, HF predictions can vary by orders of magnitude
- Why?



Ejectile Energy Distribution

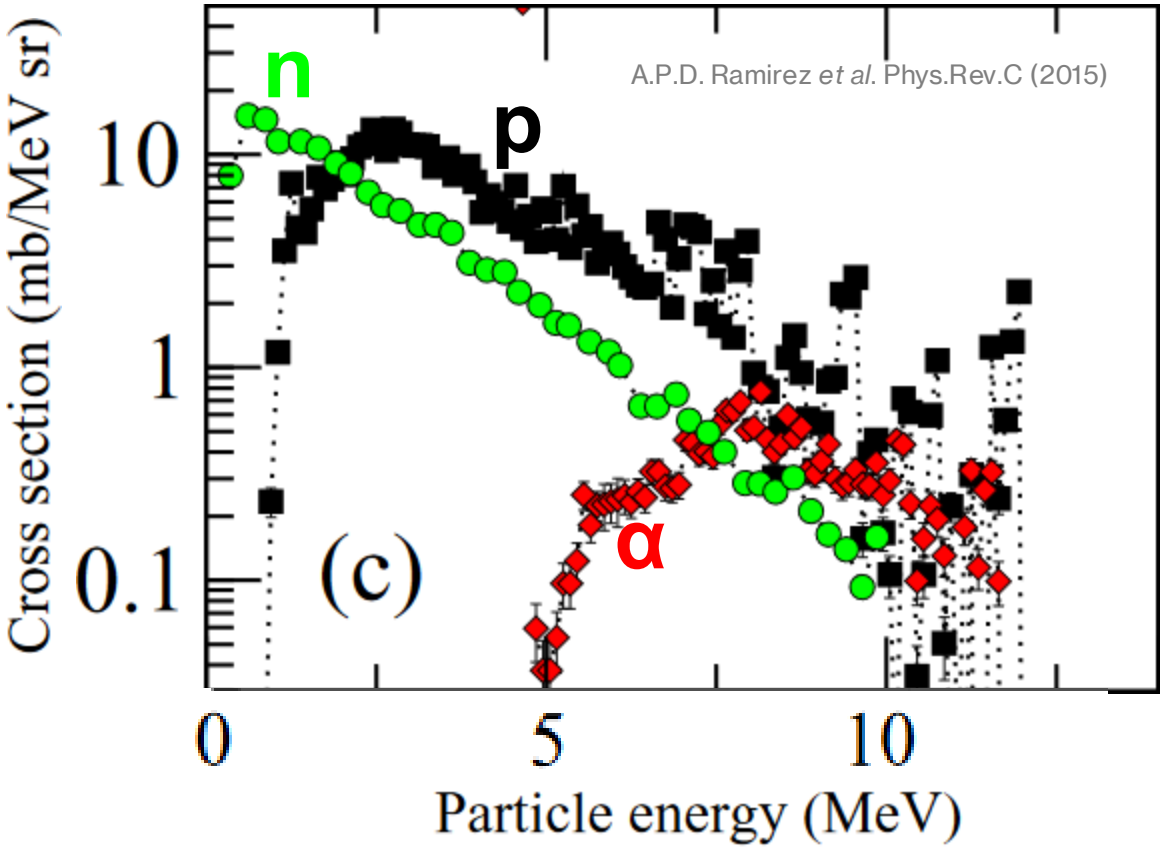
- What would the ejectile energy distribution look like?
- We've already likened this process to how molecules evaporate from a hot liquid, often called particle evaporation so we can essentially use the Maxwell-Boltzmann distribution for the ejectile energy distribution.
- The distribution is commonly written as $\boldsymbol{N(\varepsilon)d\varepsilon = \frac{\varepsilon}{T^2} e^{-\varepsilon/T} d\varepsilon}$
- Since we are using temperature here, you can see why we say this is similar to the evaporation of particles



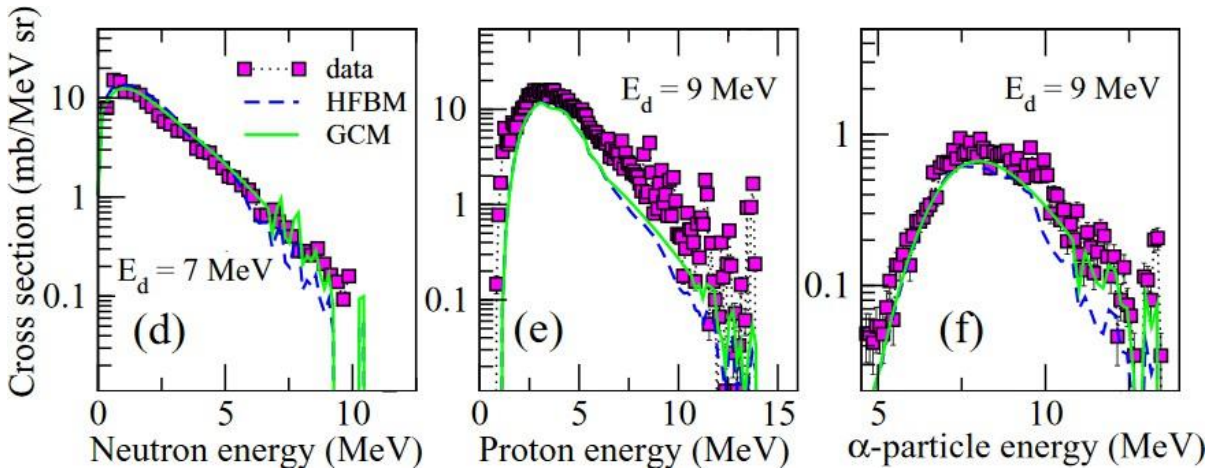


Ejectile Energy Distribution

- **Example**: ^{56}Fe + d for E_d = 7 MeV

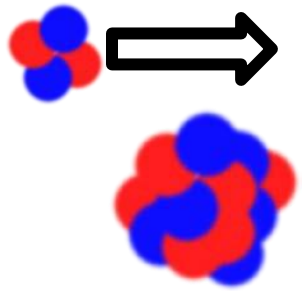


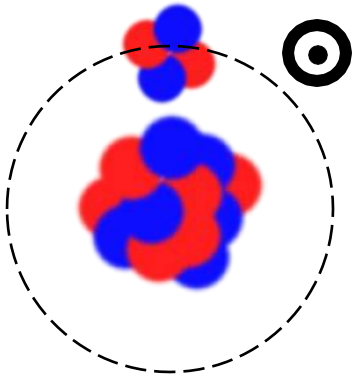
Of course, we always need to consider contributions from Coulomb, which mostly involve an overall reduction in the spectrum



Ejectile Angular Distribution

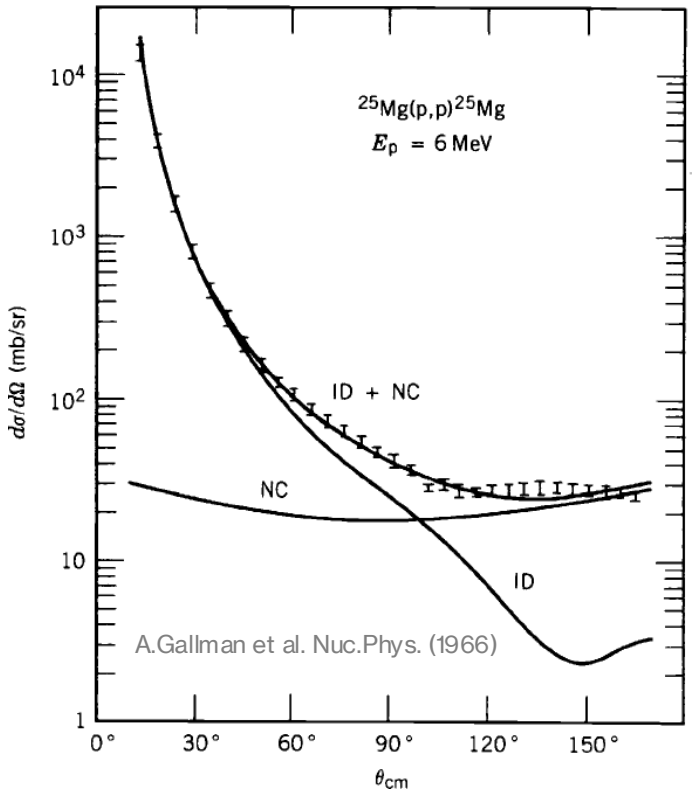
- What about the angular distribution of ejectiles?
- If we truly believe the Independence Hypothesis, then we would expect the CN to have lost all information about how it was formed.
- As such, in the center-of-mass frame we expect an isotropic emission of ejectiles.... But this is actually too naïve.
- The CN will form a range of angular momenta due to the orbital motion in the entrance channel
- This angular momenta will have to be conserved during the decay of excited states, and it will tend to focus the particles into the plane perpendicular to the emission axis and symmetric around $\theta_{cm} = 90°$

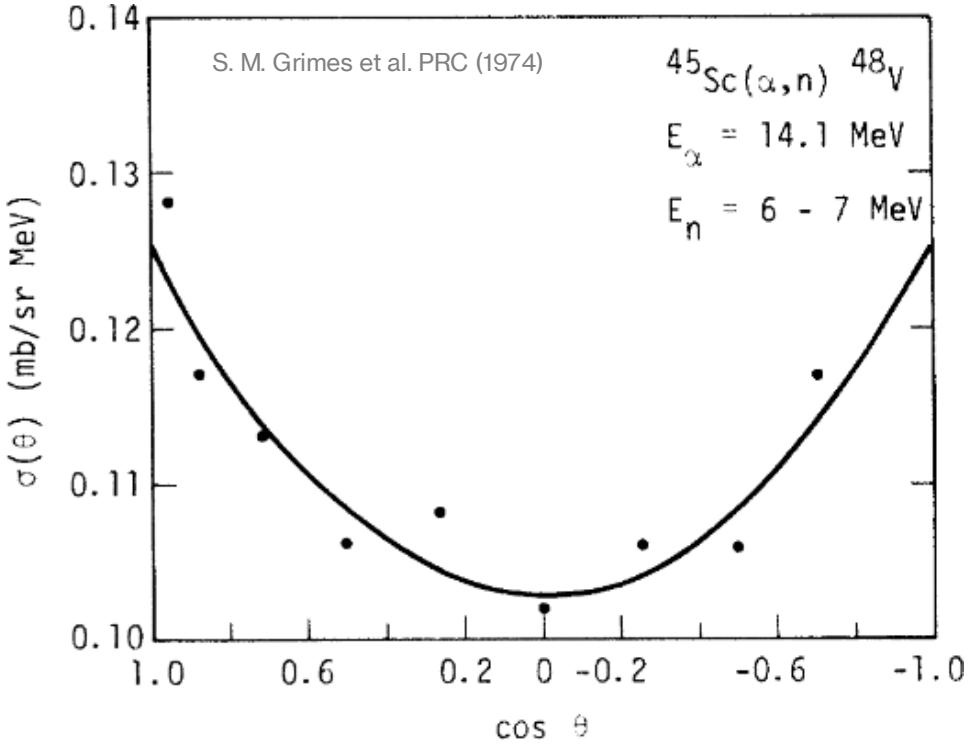




Realistic Ejectile Angular Distribution

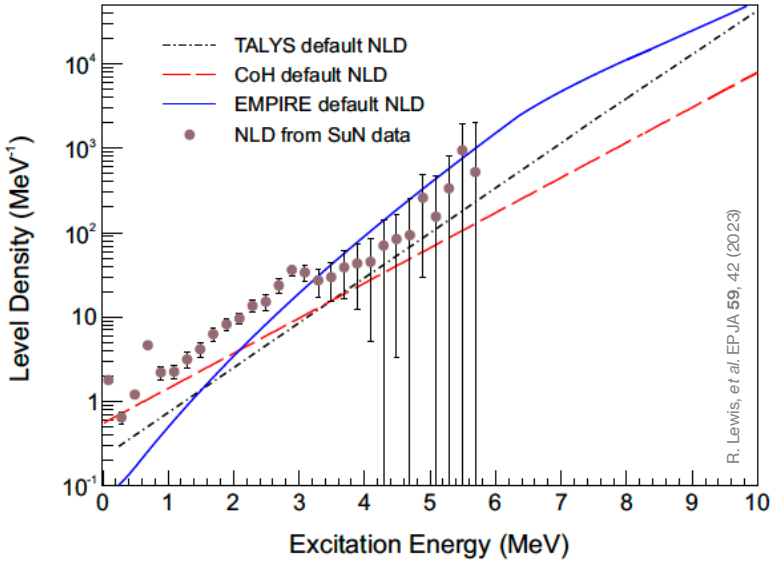
- The statistical nuclear reaction mechanism will compete with direct reactions, so backward angles are where one looks for information about the compound nucleus
- Similarly, lower bombarding energies and channels that would require multiple nucleon-transfer for a direct reaction are more promising for statistical nuclear reaction signatures





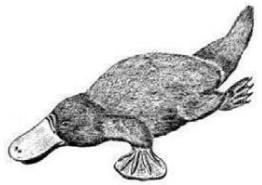
Practical Hauser Feshbach Calculations

- In general, no one is writing their own HF codes (usually)
- There are great open-source tools such as TALYS or EMPIRE that are very widely used
- There are even online versions where you can do ”quick” calculations: TALYSworld
- All of these codes require careful attention to input parameters to produce realistic, correct, and accurate calculations



TALYS-1.8

A nuclear reaction program



User Manual

Arjan Koning
Stephane Hilaire
Stephane Goriely

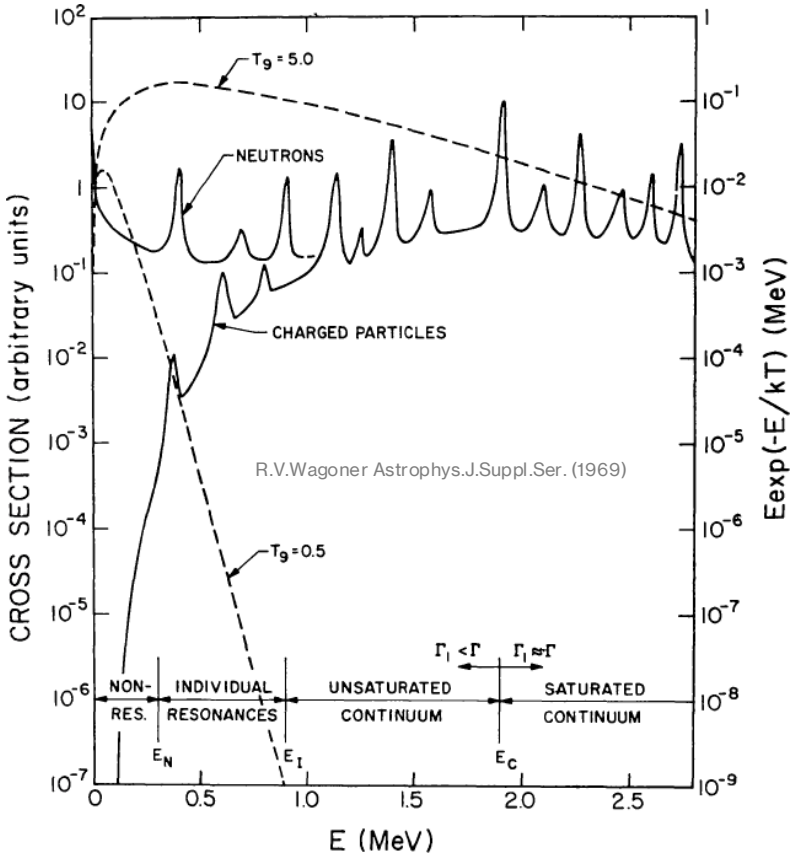
EMPIRE-II

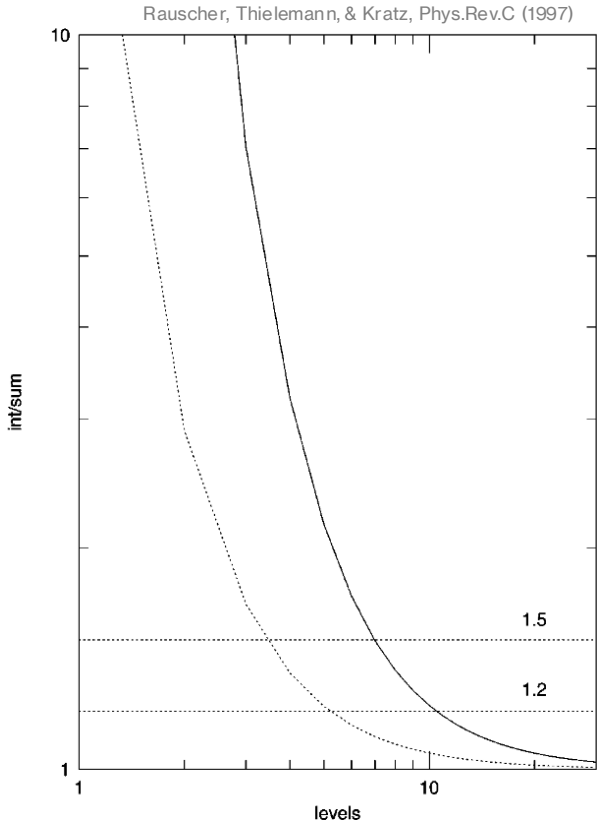
Developers

- **M. Herman (BNL), co-ordinator**
- R. Capote (IAEA)
- P. Oblozinsky (BNL)
- A. Trkov (IAEA)

Applicability of the Statistical Approach

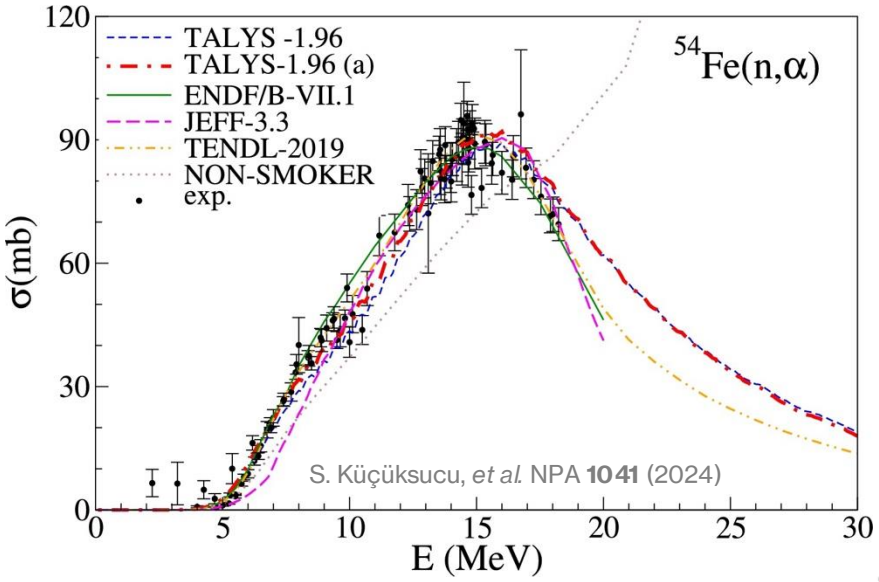
- When is it appropriate to use the Hauser-Feshbach approach vs. the Breit-Wigner sum over resonances?
- Fortunately, those that came before us have already worked this out and determined that $\gtrsim 10$ levels/Mev is sufficient

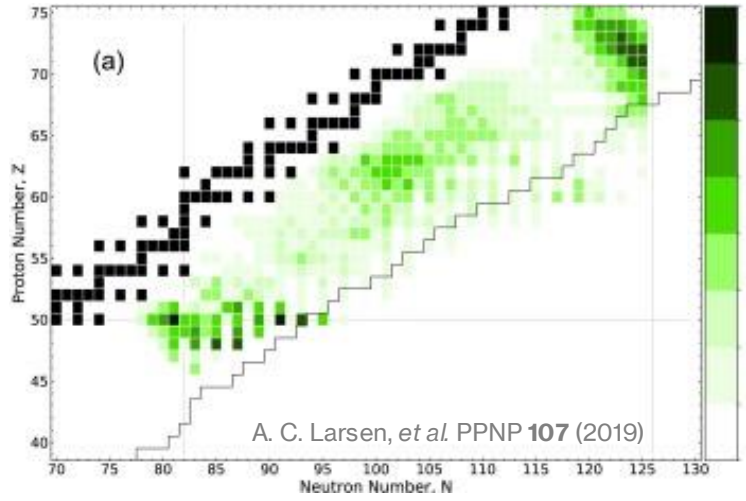


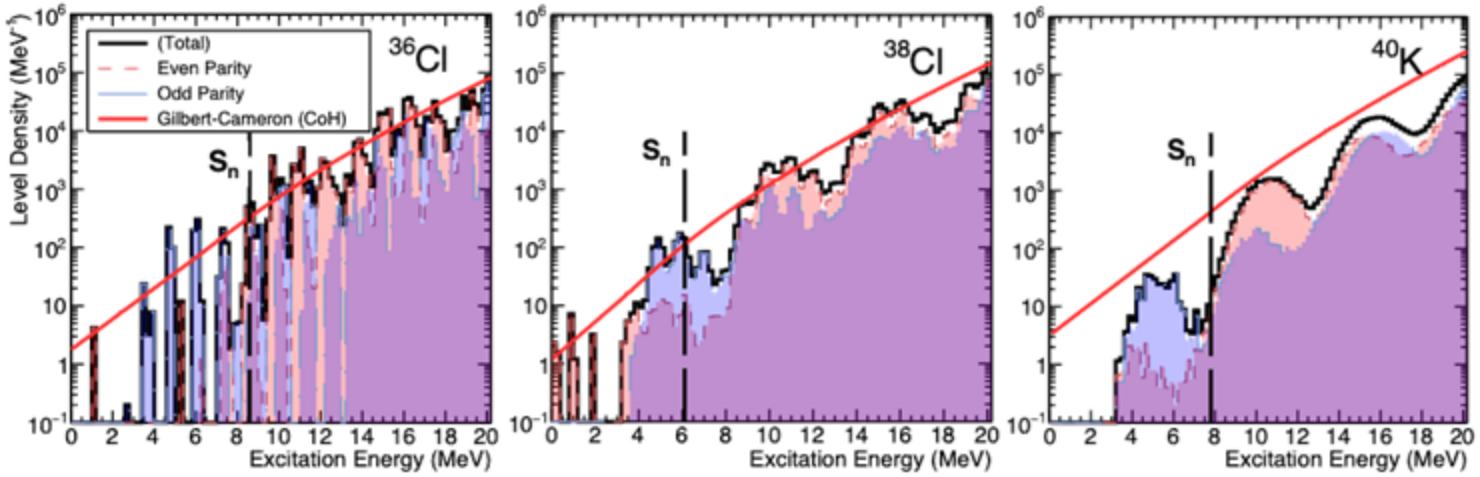


Why should you care?

- HF formalism is used to calculate medical isotope production cross sections, reactions networks from fission, nuclear astrophysics, etc.





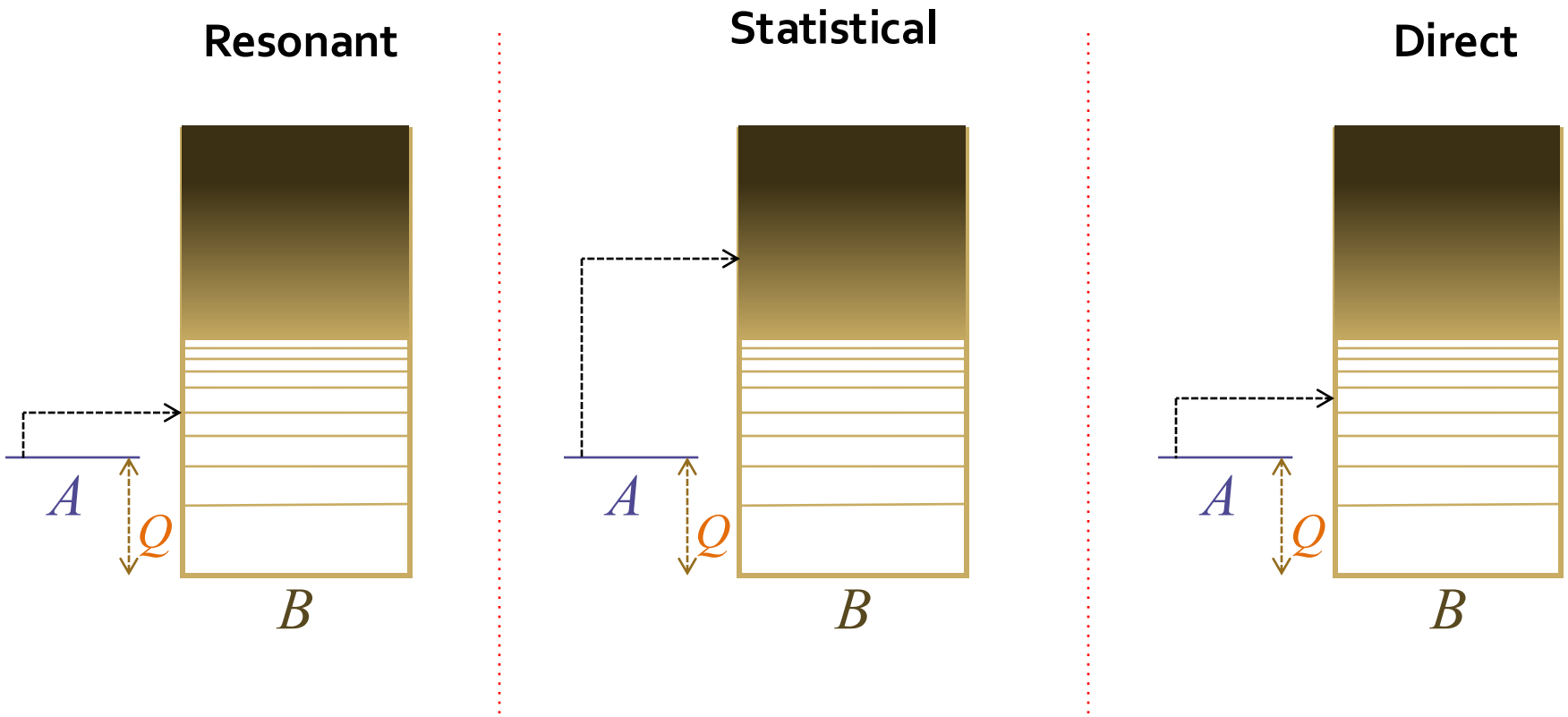


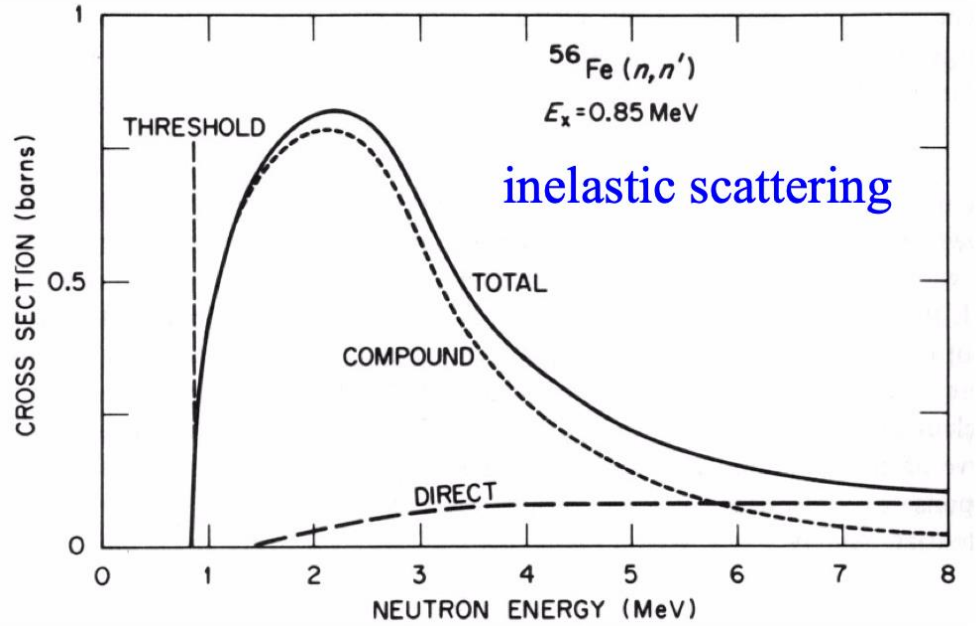
K. Hanelseman, *et al.,* EPJ Web of Conferences **322**, 02005 (2025)

Just to put things in context...

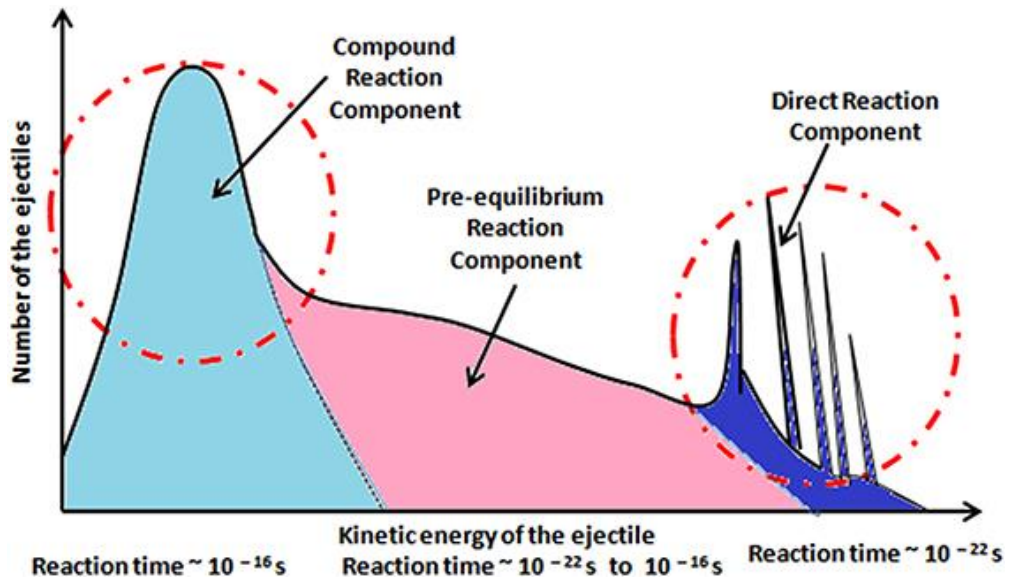
Incoming channel

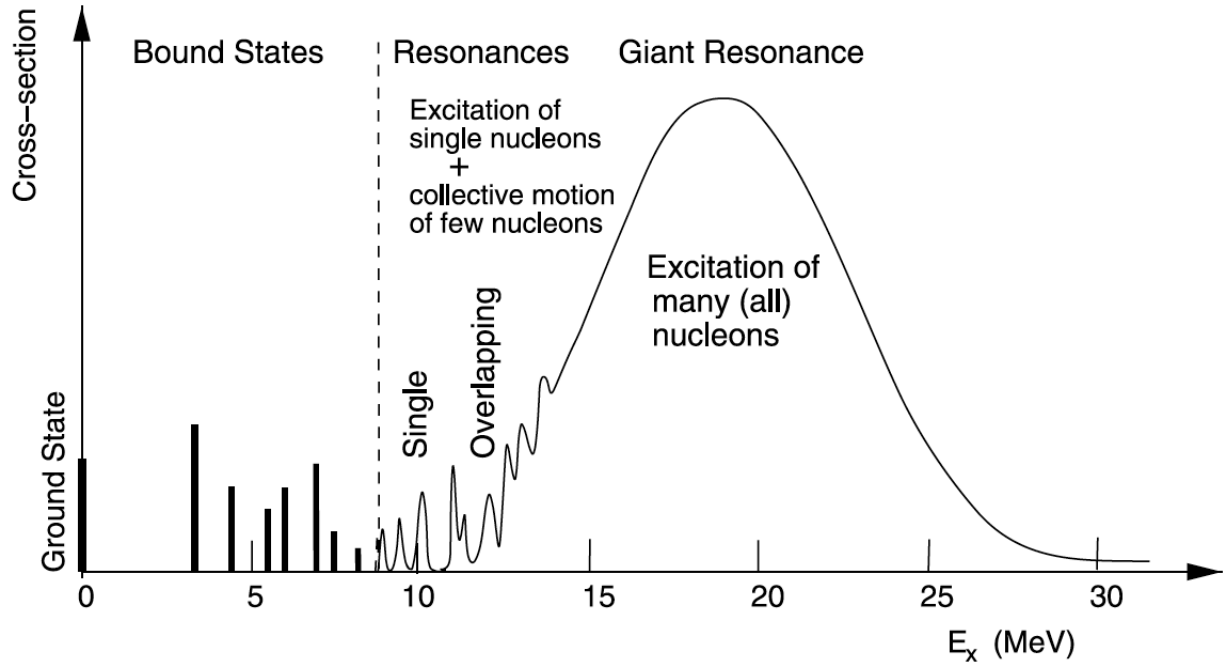
$$a + A \square B^{*} \square B + \gamma$$





inelastic scattering





Wrapping up!



Office Hours Friday
10AM, Edwards 204



Next class: Other Reaction Types



Questions?

Final Exam Topics

- You should pick a topic from the following list (or propose something to me). It cannot be closely related to your research. E-mail me a proposed topic with a list of resources you've found by next Friday at 5PM. Topics must be approved by me.

- **Islands of Inversion**
- **Shape Coexistence -- Aashish**
- **Non-axial deformation**
- **The N=126 Magic Number**
- **Superheavy Nuclei – Noah (Z=126)**
- **Fusion Reactors**
- **Reactor Neutrino Anomaly**
- **Neutrinoless double beta-decay**
- **Beta-Delayed Particle Emission -- José**
- **s-process nucleosynthesis -- Amadie**
- **r-process nucleosynthesis – Ben**
- **rp-process nucleosynthesis**
- **Nuclear Pasta in Neutron Stars -- Nick**
- **Halo Nuclei -- Mark**
- **Nuclei and Medical Imaging -- Alan**
- **Radiation Therapy Techniques**
- **Medical Isotope Production -- Unish**
- **Effective Field Theory**
- **Rare Isotope Harvesting -- Cade**
- **Neutron Imaging -- Sushma**
- **Astromers**
- **Nuclear Masses and Astrophysics**
- **Nuclear Materials Analysis**
- **Density Functional Theories- -- Parker**
- **Alpha Clustering in Nuclei**

Due Dates:
10/31 – Topic and at least 5 resources
11/18 – Outline of paper
11/25 – First Draft of Slides

What's due on Friday?

- One page document sent via e-mail

Name:

Topic:

List of 5 resources:

1.
2.
3.
4.
5.

Brief explanation of how you plan to use these resources and what aspect of the topic you plan to discuss (~ 3 sentences)